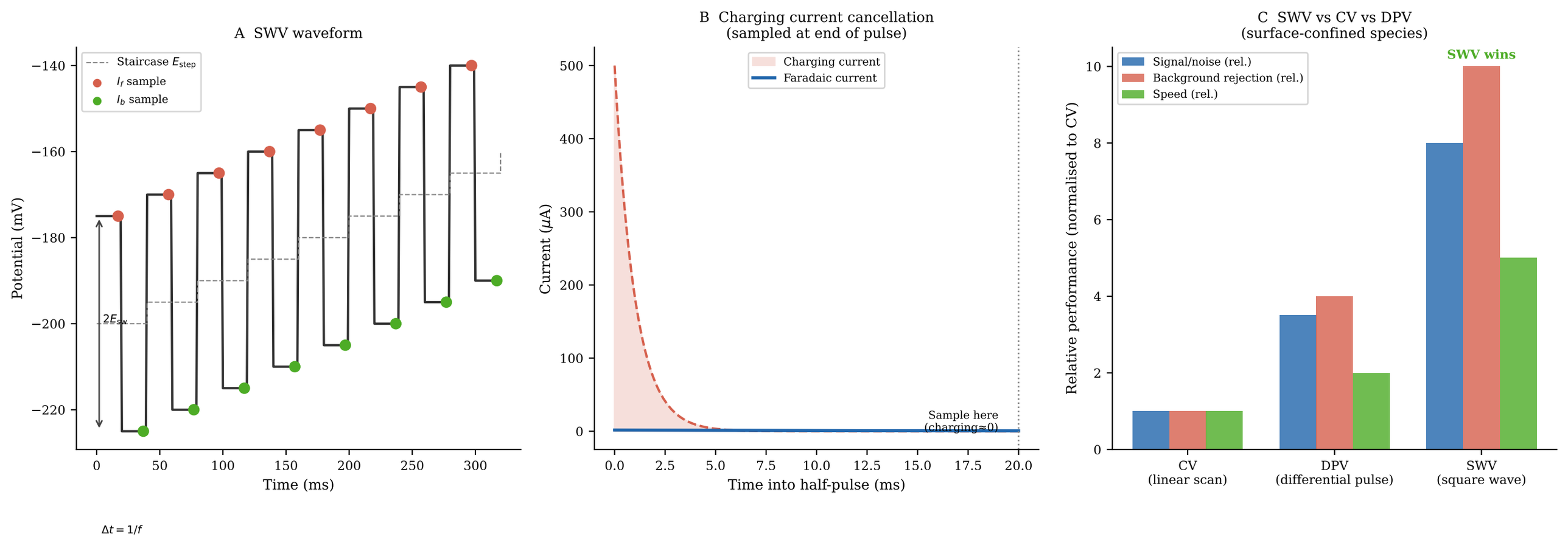
A SWV waveform
Staircase $E_{step}$
$I_f$ sample
$I_b$ sample
$2E_{sw}$
Potential (mV)
Time (ms)
$\Delta t = 1/f$
B Charging current cancellation
(sampled at end of pulse)
Charging current
Faradaic current
Sample here
(charging≈0)
Current (μA)
Time into half-pulse (ms)
C SWV vs CV vs DPV
(surface-confined species)
Signal/noise (rel.)
Background rejection (rel.)
Speed (rel.)
SWV wins
Relative performance (normalised to CV)
CV
(linear scan)
DPV
(differential pulse)
SWV
(square wave)